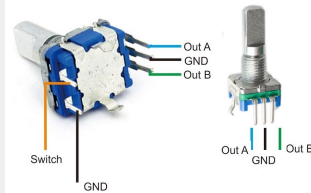


#### SD CARD

|      |         |
|------|---------|
| CS   | GP5 (7) |
| SCK  | GP2 (4) |
| MOSI | GP3 (5) |
| MISO | GP4 (6) |
| VCC  |         |
| GND  |         |



#### EC11 ENCODERS (Rotation)

|            |       |           |
|------------|-------|-----------|
| Encoder 1  | OUT A | GP6 (9)   |
| Encoder 2  | OUT A | GP7 (10)  |
| Encoder 3  | OUT A | GP8 (11)  |
| Encoder 4  | OUT A | GP9 (12)  |
| Encoder 5  | OUT A | GP10 (14) |
| Encoder 6  | OUT A | GP11 (15) |
| Encoder 7  | OUT A | GP12 (16) |
| Encoder 8  | OUT A | GP13 (17) |
| Encoder 9  | OUT A | GP14 (19) |
| Encoder 10 | OUT A | GP15 (20) |

#### EC11 ENCODERS (Push)

|              |        |           |
|--------------|--------|-----------|
| All Encoders | SWITCH | GP16 (21) |
|--------------|--------|-----------|

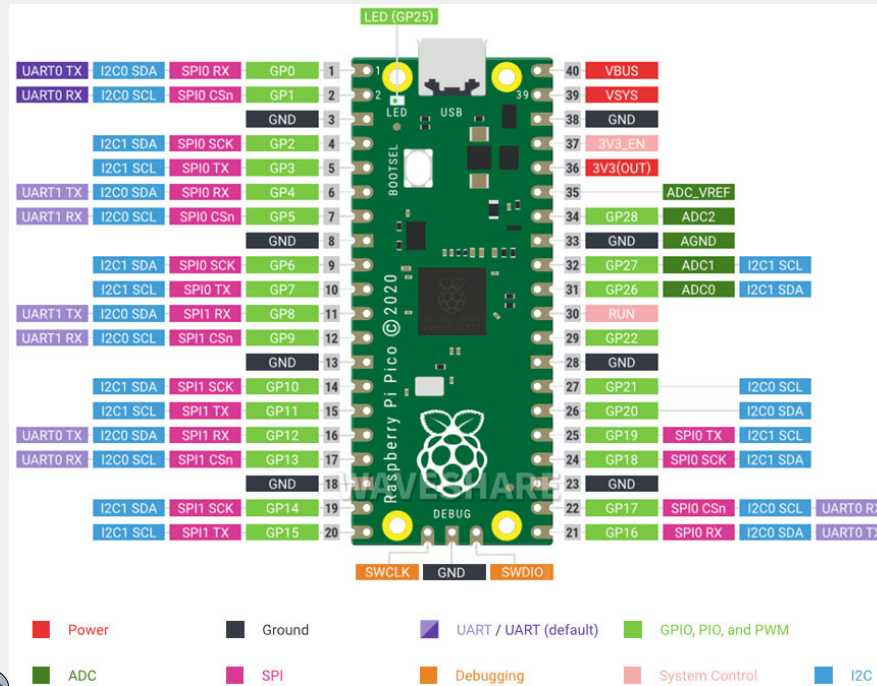
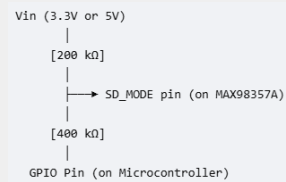
#### AUDIO - I2S (Shared by 2x MAX98357A)

|      |           |
|------|-----------|
| LRC  | GP19 (25) |
| BCLK | GP20 (26) |
| DIN  | GP21 (27) |
| GAIN | -         |
| SD   |           |
| GND  |           |
| VIN  |           |

#### SPEAKER SELECTION (SD pin)

|           |                               |
|-----------|-------------------------------|
| GP17 (22) | SPEAKER ENABLE (MAX98357A #1) |
| GP18 (24) | HANDSET ENABLE (MAX98357A #2) |

\* may need two-resistor voltage divider or transistor or something



#### BUTTONS/KEYPAD

|          |           |
|----------|-----------|
| PLUS     | GP22 (29) |
| MINUS    | GP26 (31) |
| MULTIPLY | GP27 (32) |
| DIVIDE   | GP28 (34) |

|         |         |
|---------|---------|
| HANDSET | GP0 (1) |
|---------|---------|

|     |         |
|-----|---------|
| LED | GP1 (2) |
|-----|---------|

| SD_MODE Voltage | Function                |
|-----------------|-------------------------|
| < -0.16V        | Shutdown                |
| -0.16V - 0.77V  | (Left + Right) / 2 Mono |
| -0.77V - 1.4V   | Right                   |
| >-1.4V          | Left                    |